CYT_THEORY ASSIGNMENT - 2

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Compare Permissions on Regular Files & Directories

Permission Testing Table:

Permission	Regular File (sample.txt) - Test	Observation	Directory (sample_dir) - Test	Observation
400 (r--)	Read and modify	**Read:** Yes. **Modify:** No	List and enter	**List:** No. **Enter:** No.
200 (-w-)	Read and modify	**Read:** No. **Modify:** Yes.	List, enter, create	**List:** No. **Enter:** No. **Create:** No, because x is absent.
100 (--x)	Read and execute	**Read:** No. **Execute:** Yes.	List and enter	**List:** No. **Enter:** Yes.
500 (r-x)	Read and modify	**Read:** Yes. **Modify:** No. **Execute:** Yes.	List and enter	**List:** Yes. **Enter:** Yes.
600 (rw-)	Read and modify	**Read:** Yes. **Modify:** Yes.	List, enter, create	**List:** No. **Enter:** No. **Create:** No.
700 (rwx)	Read, modify, execute	**Read:** Yes. **Modify:** Yes. **Execute:** Yes.	List, enter, create, delete	**List:** Yes. **Enter:** Yes. **Create/Delete:** Yes.

Questions and Answers

1. What does read (r) permission allow you to do with a regular file?

-- It allows you to **view/read the contents** of the file.

2. What does read (r) permission allow you to do with a directory?

-- It allows you to **list the names of files and directories** contained within it.

3. What does write (w) permission mean for a regular file?

-- It allows you to **modify or change the contents** of the file.

4. What does write (w) permission mean for a directory?

-- It allows you to **create, delete, and rename directory entries**, subject to the directory's other permissions and any special restrictions.

5. What does execute (x) permission mean for a regular file?

-- It allows the file to be **executed as a program or script**, provided the file is actually executable in a meaningful way.

6. Why is execute (x) permission important for accessing a directory?

-- For a directory, x means **search/traverse permission**. It allows you to enter the directory with cd and access items within it when the relevant permissions are also satisfied.

7. Which permission combination allows the owner to fully access a regular file?

-- 700 (rwx) — read, write, and execute.

8. Which permission combination allows the owner to fully access a directory?

-- 700 (rwx) — list contents, enter/traverse, and create/delete/rename entries.

9. Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

-- The meaning of permissions depends on the type of filesystem object. For a **regular file**, r controls reading contents, w controls modifying contents, and x controls execution. For a **directory**, r controls listing its entries, w controls creating/deleting/renaming entries, and x controls entering or traversing the directory. Therefore, the same permission letters provide different capabilities depending on whether the object is a file or directory.